

DIGITAL IMAGE PROCESSING

LECTURE # 8

IMAGE ENHANCEMENT IN SPATIAL DOMAIN-IV

Contact Information

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- Gradient Operators
 - ❑ Noise Factor
 - ❑ Blurred Edges
 - ❑ Optimal Edge Detector
 - ❑ Canny Edge Detection

Noise in an image

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❑ Problem with Edge Localization

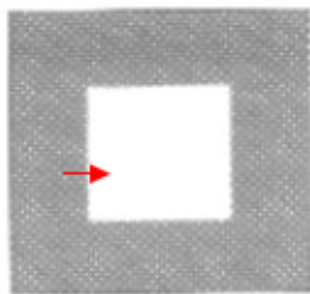
- ✓ High Threshold may suppress meaningful edges
- ✓ Low Threshold may include unwanted edges
- ✓ Noise may have high magnitude

Text

Blurred Edges

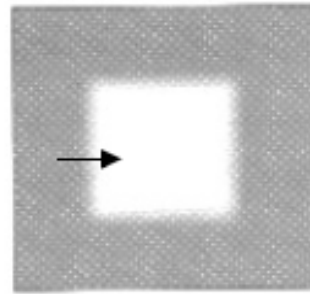
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- The detected edge can be rather broad in the case of diffuse edges, resulting in a thick band of pixels instead of a single point of maximum gradient.



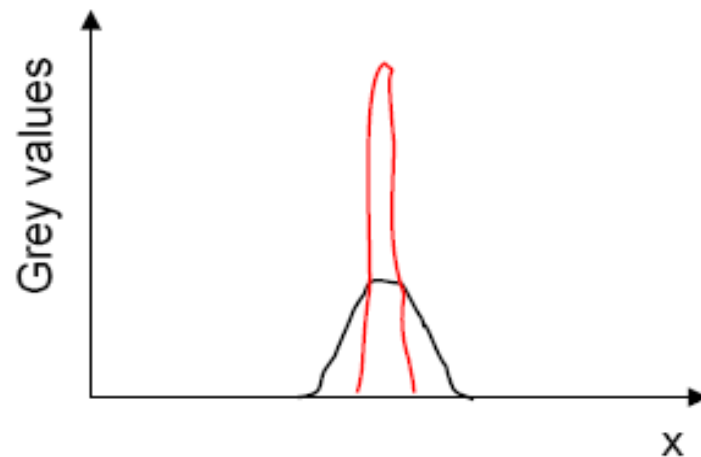
(a)

Slightly blurred



(b)

Heavily blurred



Canny Edge Detection

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- ❑ The **Canny** edge detection operator was developed by John F. Canny in 1986 and uses a multi-stage algorithm to detect a wide range of edges in images.
- ❑ Canny's aim was to discover the optimal edge detection algorithm. In this situation, an "optimal" edge detector means:
 - ✓ *good detection* – the algorithm should mark as many real edges in the image as possible.
 - ✓ *good localization* – edges marked should be as close as possible to the edge in the real image.
 - ✓ *minimal response* – a given edge in the image should only be marked once, and where possible, image noise should not create false edges.

Text

Canny Edge Detection

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▪ Algorithm Steps

- Image smoothing
- Gradient computation
- Edge direction computation
- Non-maximum suppression
- Hysteresis Thresholding

Image Smoothing

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- ❑ Reduces image noise that can lead to erroneous output
- ❑ Performed by convolution of the input image with a Smoothing filter

1 — 159	2	4	5	4	2
	4	9	12	9	4
	5	12	15	12	5
	4	9	12	9	4
	2	4	5	4	2

Image Smoothing

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Gradient Computation

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- ❑ Determines intensity changes
- ❑ High intensity changes indicate edges
- ❑ Performed by convolution of smoothed image with masks to determine horizontal and vertical derivatives

-1	0	1
-2	0	2
-1	0	1

∇_x

-1	-2	-1
0	0	0
1	2	1

∇_y

Gradient Computation

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- Gradient magnitude determined by adding X and Y gradient images

$$|\nabla| = |\nabla_x| + |\nabla_y|$$



Edge Direction Computation

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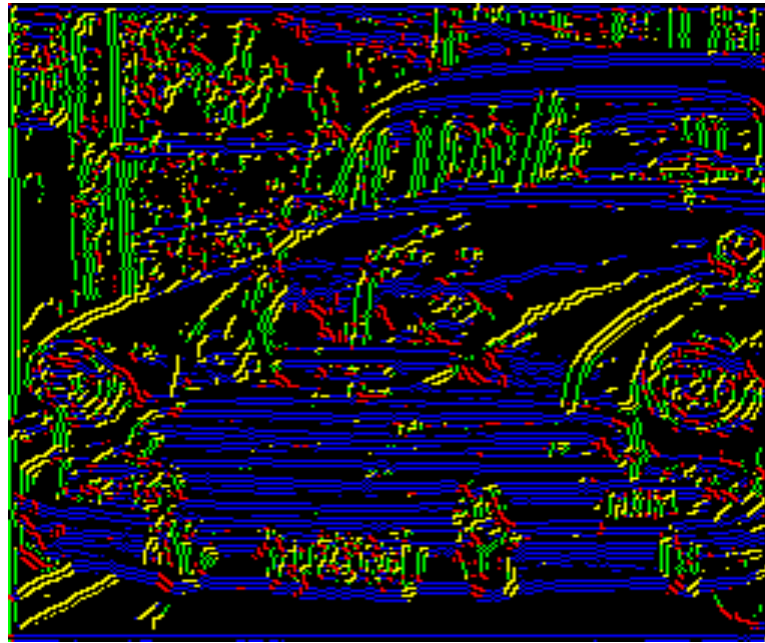
- Edge directions are determined from running a computation on the X and Y gradient images

$$\Theta_{x,y} = \tan^{-1} \frac{\nabla_x}{\nabla_y}$$

- Edge directions are then classified by their nearest 45° angle

Edge Direction Computation

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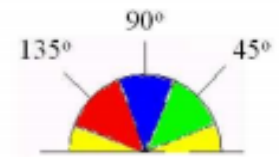
Compute θ' by rounding the angle θ to one of four directions 0° , 45° , 90° , or 135° . Obviously for edges, $180^\circ = 0^\circ$, $225^\circ = 45^\circ$, etc. This means θ in the ranges $[-22.5^\circ \text{ to } 22.5^\circ]$ and $[157.5^\circ \text{ to } 202.5^\circ]$ would “round” to $\theta' = 0^\circ$. For a pictorial representation, each edge take on one of four colors: Here, the colors would repeat on the lower half of the circle (green around 225° , blue around 270° , and red around 315°)

0°

90°

45°

135°

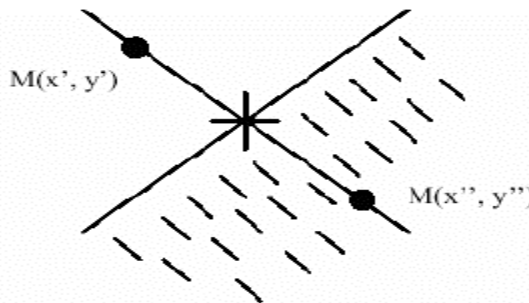


Non-Maximum Suppression

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- Non-maximum suppression is an edge thinning technique.
- Given estimates of the image gradients, a search is then carried out to determine if the gradient magnitude assumes a local maximum in the gradient direction.
- This is worked out by passing a 3x3 grid over the intensity map.
- From this stage referred to as non-maximum suppression, a set of edge points, in the form of a binary image, is obtained.

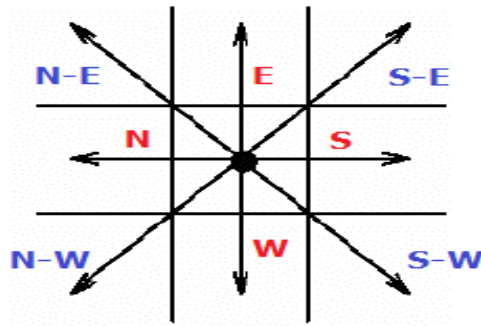
$$M(x, y) = \begin{cases} M(x, y) & \text{if } M(x, y) > M(x', y') \text{ \& } \\ & \text{if } M(x, y) > M(x'', y'') \\ 0 & \text{otherwise} \end{cases}$$



Non-Maximum Suppression

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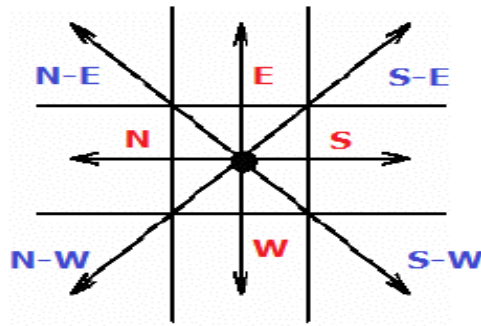
- At every pixel, it suppresses the edge strength of the center pixel (by setting its value to 0) if its magnitude is not greater than the magnitude of the two neighbors in the gradient direction.
 - if the rounded angle is zero degrees the point will be considered to be on the edge if its intensity is greater than the intensities in the **west and east** directions,
 - if the rounded angle is 90 degrees the point will be considered to be on the edge if its intensity is greater than the intensities in the **north and south** directions,
 - if the rounded angle is 135 degrees the point will be considered to be on the edge if its intensity is greater than the intensities in the **north west and south east** directions,
 - if the rounded angle is 45 degrees the point will be considered to be on the edge if its intensity is greater than the intensities in the **north east and south west** directions.



Non-Maximum Suppression

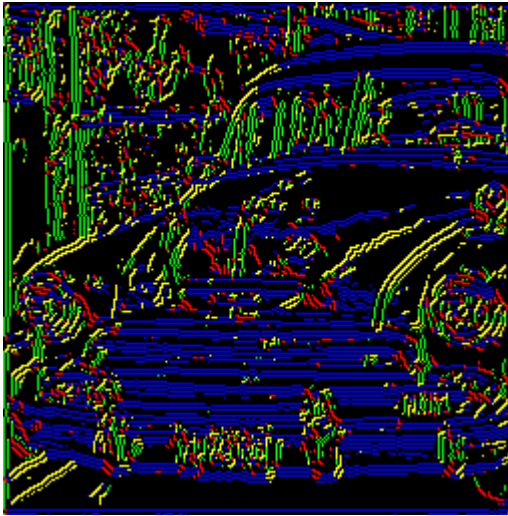
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- If pixel (x, y) has the highest gradient magnitude of the three pixels examined, it is kept as an edge.
- If one of the other two pixels has a higher gradient magnitude, then pixel (x, y) is not on the “center” of the edge and should not be classified as an edge pixel.
- The “non-maximal suppression” step keeps only those pixels on an edge with the highest gradient magnitude.
- These maximal magnitudes should occur right at the edge boundary, and the gradient magnitude should fall off with distance from the edge.



Non-Maximum Suppression

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Thresholding

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- Reduce number of false edges by applying a threshold T
 - ❑ all values below T are changed to 0
 - ❑ selecting a good values for T is difficult
 - ❑ some false edges will remain if T is too low
 - ❑ some edges will disappear if T is too high

Thresholds

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- Thresholds are important, done before or during edge detection.



original image



very high threshold

Thresholds

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original image



very high threshold



Thresholds

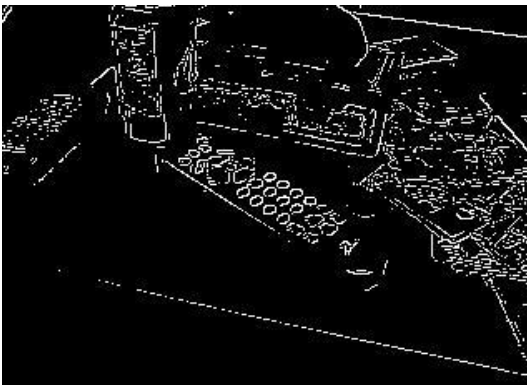
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original image



very high threshold →



Thresholds

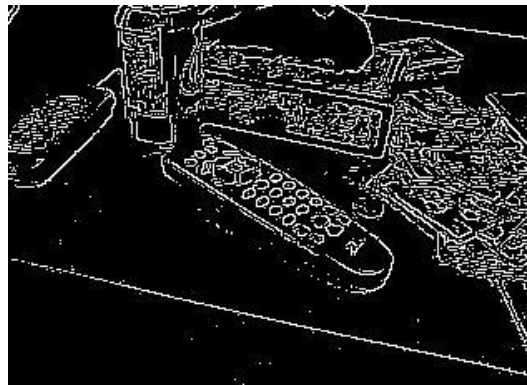
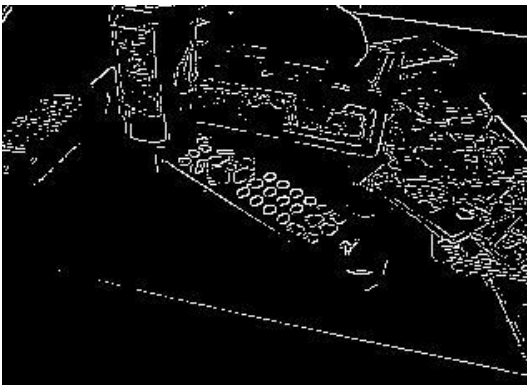
22



original image



very high threshold



reasonable

Thresholds

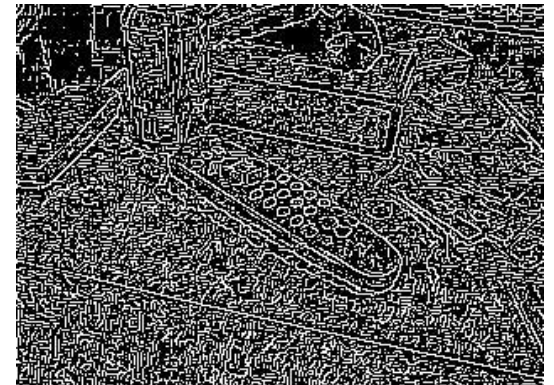
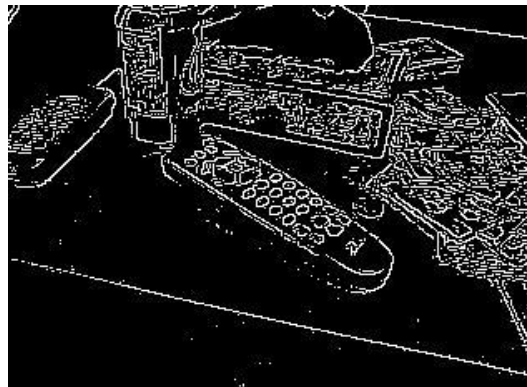
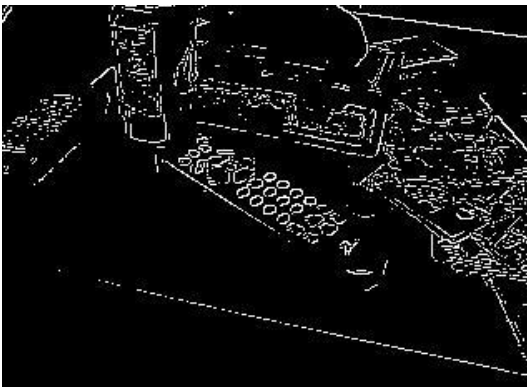
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original image



very high threshold



reasonable

too low !

this all takes time...

Thresholding

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- ❑ Thresholding with hysteresis requires two thresholds - high and low.
- ❑ We begin by applying a high threshold. This marks out the edges we can be fairly sure are genuine. Starting from these, using the directional information derived earlier, edges can be traced through the image.
- ❑ While tracing an edge, we apply the lower threshold, allowing us to trace faint sections of edges.
- ❑ Once this process is complete, we have a binary image where each pixel is marked as either an edge pixel or a non-edge pixel.

Hysteresis Thresholding

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- Apply two thresholds in the suppressed image

□ $T_2 = 2T_1$

- two images in the output



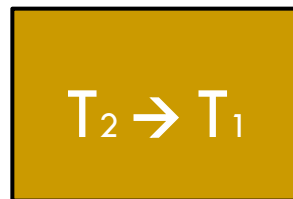
↑
 T_H

Gives Strong Edge pixels



↑
 T_L

Gives Weak and Strong Edge pixels



Hysteresis Thresholding

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- ❑ the image from T_2 contains fewer edges but has gaps in the contours
- ❑ the image from T_1 has many false edges
- ❑ **combine** the results from T_1 and T_2
- ❑ link the edges of T_2 into contours until we reach a gap
- ❑ link the edge from T_2 with edge pixels from a T_1 contour until a T_2 edge is found again

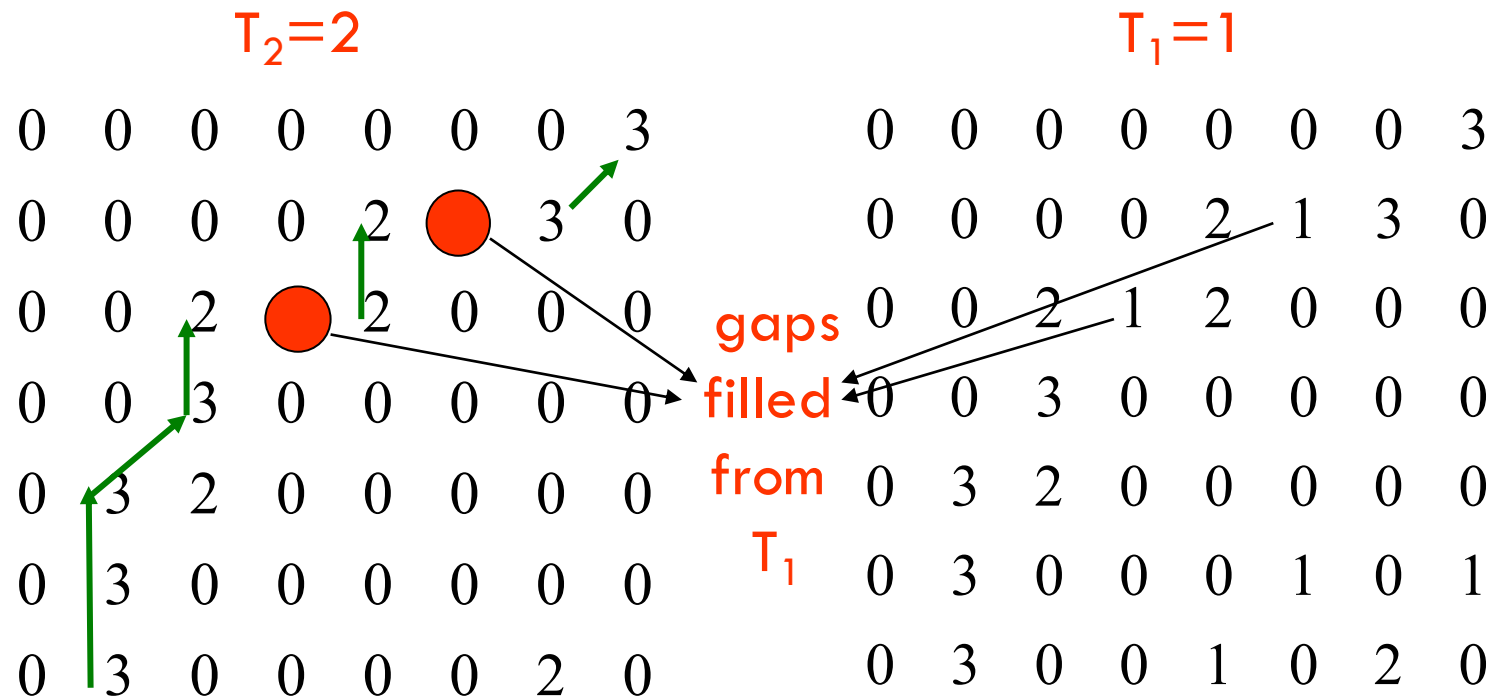
Hysteresis Thresholding

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- ❑ If pixel (x, y) has gradient magnitude less than T_L discard the edge (write out black).
- ❑ If pixel (x, y) has gradient magnitude greater than T_H keep the edge (write out white).
- ❑ If pixel (x, y) has gradient magnitude between T_L and T_H and any of its neighbors in a 3×3 region around it have gradient magnitudes greater than T_H , keep the edge (write out white).
- ❑ If none of pixel (x, y) 's neighbors have high gradient magnitudes but at least one falls between T_L and T_H , search the 5×5 region to see if any of these pixels have a magnitude greater than T_H . If so, keep the edge (write out white).
- ❑ Else, discard the edge (write out black).

Hysteresis Thresholding

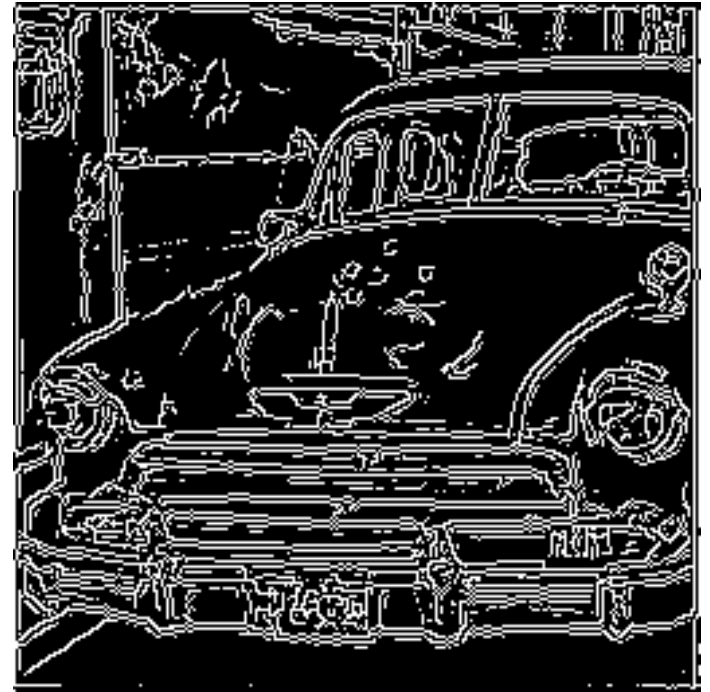
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- A T_2 contour has pixels along the green arrows
- **Linking:** search in a 3x3 of each pixel and connect the pixel at the center with the one having greater value
- Search in the direction of the edge

Hysteresis Thresholding

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References

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1. DIP by Gonzalez

For any query Feel Free to ask



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